CHAPTER I

INTRODUCTION

Public service delivery is one of the most important responsibilities of local government units (LGUs), as it directly affects the daily lives of citizens. Services such as permit processing, document requests, emergency assistance, public advisories, and community issue reporting must be accessible, organized, and responsive (Palma et al., 2023). As technology continues to develop, digital platforms have become essential tools for improving the speed, transparency, and accessibility of government services (United Nations Department of Economic and Social Affairs [UN DESA], 2024). In the field of Information Technology, capstone projects are expected to provide practical solutions to real-world problems by applying technical knowledge, system development, and problem-solving skills.

Despite the increasing use of digital systems, many LGUs still rely on manual or fragmented processes. Citizens often need to visit different offices, search for service requirements, wait for updates, or use separate communication channels to complete government transactions. This can result in delays, confusion, inefficient communication, and limited transparency (Sigwejo & Pather, 2016). These challenges show the need for a centralized platform that can help citizens access local government services more conveniently while helping LGUs manage requests and reports more efficiently (Sabani et al., 2023).

The Automated Governance and Public Service Platform (AGAPP) is a proposed digital solution designed to modernize and streamline the delivery of government services within local government units. Inspired by systems such as the MyNaga Mobile Application, AGAPP aims to serve as a centralized and user-friendly platform that improves accessibility, transparency, and efficiency in public service delivery (Mergel et al., 2019). The system includes major features such as an e-services portal, citizen guide system, emergency assistance access, news and updates, issue reporting, request tracking, community forum, chatbot customer support, and interactive maps.

Through AGAPP, citizens will be able to access government services, submit reports, track requests, receive announcements, and locate important offices and landmarks using a single platform. On the other hand, LGUs will be able to organize reports, monitor concerns, provide updates, and respond more effectively to community needs. Therefore, this project is designed to support digital transformation in local governance by improving service delivery, citizen engagement, and communication between the government and the community (Furtado et al., 2023).

Project Context

Local government units are responsible for providing essential public services to citizens. These services include processing permits, issuing official documents, responding to emergencies, addressing community concerns, and providing public information. However, when these services are handled manually or through disconnected systems, citizens may experience inconvenience and delays (Bales et al., 2024). They may also find it difficult to know the correct procedures, requirements, schedules, or contact information for specific government services (Mascara, 2025).

At present, many local government services are still limited by traditional processes and fragmented communication channels. Citizens may need to physically visit municipal offices to inquire, submit documents, or follow up on requests (Aminah & Saksono, 2021). In some cases, public concerns such as damaged utility poles, clogged drainage systems, potholes, stray animals, and lost or found items may not be reported efficiently because there is no centralized reporting platform. These issues may lead to slower response times and reduced public satisfaction (Sagarik et al., 2018).

The AGAPP project addresses these problems by proposing a centralized platform that integrates multiple government services into one system. Based on the project reference, AGAPP is designed to provide centralized access to e-services, complete service directories, emergency hotlines, official announcements, service requests, issue reports, request tracking, community discussions, chatbot support, and digital maps (Organisation for Economic Co-operation and Development [OECD], 2020).

For the purposes of this capstone project, the Municipality of Liliw, Province of Laguna has been identified as the pilot local government unit through which AGAPP will be developed and evaluated. While the pilot is conducted in a single municipality, the platform is designed from the start to be multi-LGU capable: each participating LGU is provisioned in the system with strict data isolation. A platform-level Super Admin role is responsible for provisioning new LGUs, monitoring cross-LGU analytics, and supervising compliance artifacts under the Data Privacy Act (Republic of the Philippines, 2018). This design allows AGAPP to be adopted by other municipalities in future deployments without changes to the core system (Department of Information and Communications Technology [DICT], 2022).

The main problem addressed by this study is: How can a centralized digital platform improve service delivery, citizen engagement, and response time in local government units?

To answer this problem, AGAPP will be developed as a prototype system that brings together different public service functions into one accessible platform. The project aims to reduce the need for physical visits, minimize delays, improve communication, and promote transparency in public service delivery (Rahmadany & Ahmad, 2021). This is aligned with the purpose of a capstone project, which focuses on solving real-world problems through practical and technology-based solutions.

Project Purpose

The purpose of this study is to design and develop the Automated Governance and Public Service Platform (AGAPP), a centralized digital system that aims to improve the efficiency, accessibility, and transparency of public service delivery in local government units. The project seeks to address common problems such as fragmented service systems, slow processing times, limited access to information, and inefficient communication between citizens and government offices.

AGAPP will integrate important features such as an e-services portal, issue reporting with image capture and GPS location tagging, real-time request tracking, emergency assistance, notification and update system, community forum, chatbot support, and interactive maps. These features are intended to provide citizens with a more convenient way to access services and allow LGUs to manage public concerns more effectively.

The beneficiaries of the proposed project are the following:

To the Citizens/Residents. Citizens and residents are the primary beneficiaries of AGAPP. Through the platform, they will be able to access government services anytime and anywhere. They can process permits, request documents, report community issues, track the status of their requests, view announcements, and access emergency hotlines. This reduces the need for physical visits to government offices and helps save time, effort, and transportation costs. The system also gives citizens a more active role in community development by allowing them to report concerns and participate in discussions through the community forum.

To the Local Government Units (LGUs). Local government units will benefit from AGAPP by improving their service management and operational efficiency. The system will help LGU personnel organize service requests, monitor reports, update citizens, and respond to concerns in a more systematic way. Reports with images and GPS location data can help administrators identify the exact location of community issues, allowing faster response and better decision-making. The platform also supports transparency because users can track the progress of their submitted requests and reports.

To the Community. The community will benefit from improved public services, faster reporting of concerns, and better access to local government information. Features such as emergency assistance, news and updates, issue reporting, and interactive maps can contribute to community safety, cleanliness, awareness, and participation. The community forum also encourages citizens to share concerns, exchange information, and participate in local discussions.

To Future Researchers and Developers. Future researchers and developers may use AGAPP as a reference for studies related to e-governance, public service automation, smart city applications, system development, and community-based digital platforms. The project may also serve as a foundation for future improvements, such as full LGU deployment, database integration, advanced analytics, and mobile application development.

Project Objectives

General Objective

The general objective of this project is to design, develop, and implement the Automated Governance and Public Service Platform (AGAPP), a centralized digital system that enhances the efficiency, accessibility, and transparency of public service delivery in local government units while improving citizen engagement and response time to community concerns.

Specific Objectives

This study has the following specific objectives:

1. To analyze the current challenges and limitations in existing public service delivery systems of local government units, particularly in terms of accessibility, processing time, and communication with citizens.
2. To design a user-friendly and centralized platform interface that allows users to easily navigate and access available government services.
3. To develop an e-services portal that allows users to submit applications for permits and document requests online, generating a pre-filled PDF and a reference QR code that the citizen presents at the Municipal Hall payment counter in person to complete payment and claim the released document.
4. To create an issue reporting system with image capture and GPS location tagging for reporting community concerns such as potholes, damaged utility poles, clogged drainage systems, stray animals, missing pets, and lost or found items.
5. To develop a real-time tracking system that allows users to monitor the status of their submitted service requests and issue reports.
6. To integrate emergency assistance features that provide quick access to important hotlines and services such as police, fire protection, hospitals, and other emergency contacts.
7. To build a notification and update system for public announcements, advisories, alerts, and request status updates.

8. To implement a community forum that allows citizens to share concerns, exchange information, and participate in community discussions.
9. To develop interactive maps for municipal offices and key town landmarks to help users locate government offices, hospitals, schools, tourist spots, and other important destinations.
10. To test and evaluate the system's functionality, usability, performance, and effectiveness through system testing and user feedback.

Project Scope and Limitation

Project Scope

The Automated Governance and Public Service Platform (AGAPP) covers the design, development, and testing of a centralized digital platform for local government service delivery, composed of a citizen mobile application and two web-based administrator dashboards (LGU Admin and Super Admin). The platform will serve as a unified digital system where citizens can access local government services, submit requests, report community issues, receive updates, and communicate with the local government.

The system will include an E-Services Portal that allows users to submit applications for permits and request official documents online. The portal generates a pre-filled application PDF and a reference QR code. The citizen then visits the Municipal Hall payment counter in person, presents the QR code to complete the required payment, and claims the released document on-site. It will also include a Complete Service Directory that provides a list of available city or municipal services, including procedures, requirements, schedules, locations, and contact information. These features are intended to help citizens understand and access government services more efficiently.

AGAPP will also provide a Citizen Guide System that contains structured information about government processes. This includes categories such as ID registration and licenses, benefits and contributions, financial support, specialized assistance, and government offices. Each entry will include relevant details such as location, schedule, and contact information.

The system will include an Emergency Assistance Access feature that allows users to quickly contact emergency services, including police, fire protection, and hospitals. This feature provides multiple contact numbers for emergency services, allowing citizens to access help more conveniently during urgent situations.

Another major feature of the system is the News and Updates module, which provides centralized access to official announcements, public advisories, and local news. This feature will help citizens stay informed about important government activities and developments.

The project will also include an Issue Reporting and Service Request System. This module allows users to report community concerns directly through the platform. Reports may include potholes, damaged or non-functioning utility poles, clogged drainage systems, stray animals, missing pets, and lost or found items. Some reports will include image capture and GPS location tagging to help administrators identify the exact location of the issue. This feature supports faster communication between citizens and local authorities.

AGAPP will also include a Request and Report Tracking System. This feature allows users to monitor the status of their submitted requests and reports. The tracking system will show statuses such as Submitted, Under Review, In Progress, Resolved, and Rejected. This promotes transparency and helps users stay updated on the progress of their concerns.

The system will include User Personalization features such as light mode and dark mode to improve user experience. It will also provide External Links and Social Media Integration, allowing users to access official government pages such as Facebook and YouTube.

A Community Forum will also be included in the platform. This will serve as a space where citizens can engage in discussions, share concerns, and exchange information relevant to their community. This feature promotes civic participation and open communication between citizens and the local government.

The system will also include Customer Support through a chatbot that can answer frequently asked questions about the system or the municipality. This feature will help users get quick responses to common concerns.

Lastly, AGAPP will provide a Municipal Hall Interactive Map and a Town Map with Key Landmarks. The municipal hall map will help citizens locate specific offices and service areas, while the town map will highlight important locations such as government offices, hospitals, schools, tourist spots, and other landmarks.

The project scope also includes system testing and evaluation to assess the functionality, usability, performance, and effectiveness of AGAPP. The project will be developed as a prototype and will focus on demonstrating the core functions of the proposed system.

Project Limitations

Although AGAPP aims to provide a comprehensive digital platform for local government services, the project has several limitations.

- Prototype Phase: The system will be limited to prototype development and simulation. It will not include full deployment across all local government units. The project will focus on demonstrating the system's core features and functionality rather than implementing it as a fully operational government system.
- External Integrations: Integration with external government databases, national identification systems, and other third-party government platforms will be limited due to access restrictions, privacy concerns, and technical requirements. The system may include simulated data or prototype-level integration only.
- Connectivity Requirement: The system will require internet connectivity for most features such as submitting service requests, submitting issue reports, receiving notifications, and updating tracking statuses. Limited offline access is supported

only for read-only content that has been cached on the device during a previous session, such as the service directory.

- Hardware Dependency: The accuracy of GPS location tagging and image capture will depend on the user's device, internet connection, and location settings. Low-quality images or inaccurate location data may affect the quality of submitted reports.
- User Testing Scale: Due to time and resource constraints, system testing will be conducted with a limited number of respondents. As a result, the testing results may not fully represent all possible users of the system.
- Budget Limits: Budget limitations may restrict the use of premium APIs, cloud services, advanced cybersecurity tools, and other paid software resources.
- Future Scope: Long-term maintenance, full-scale deployment, continuous system monitoring, and future upgrades are beyond the scope of this study. These may be considered for future development.

Conceptual Model of the Project

The conceptual model of the project presents the overall structure and flow of the AGAPP system. It follows the Input-Process-Output (IPO) Model, which identifies the necessary inputs, the processes involved in system development, and the expected output of the project. The conceptual model helps explain how the system will be developed and how its components work together to achieve the project objectives.

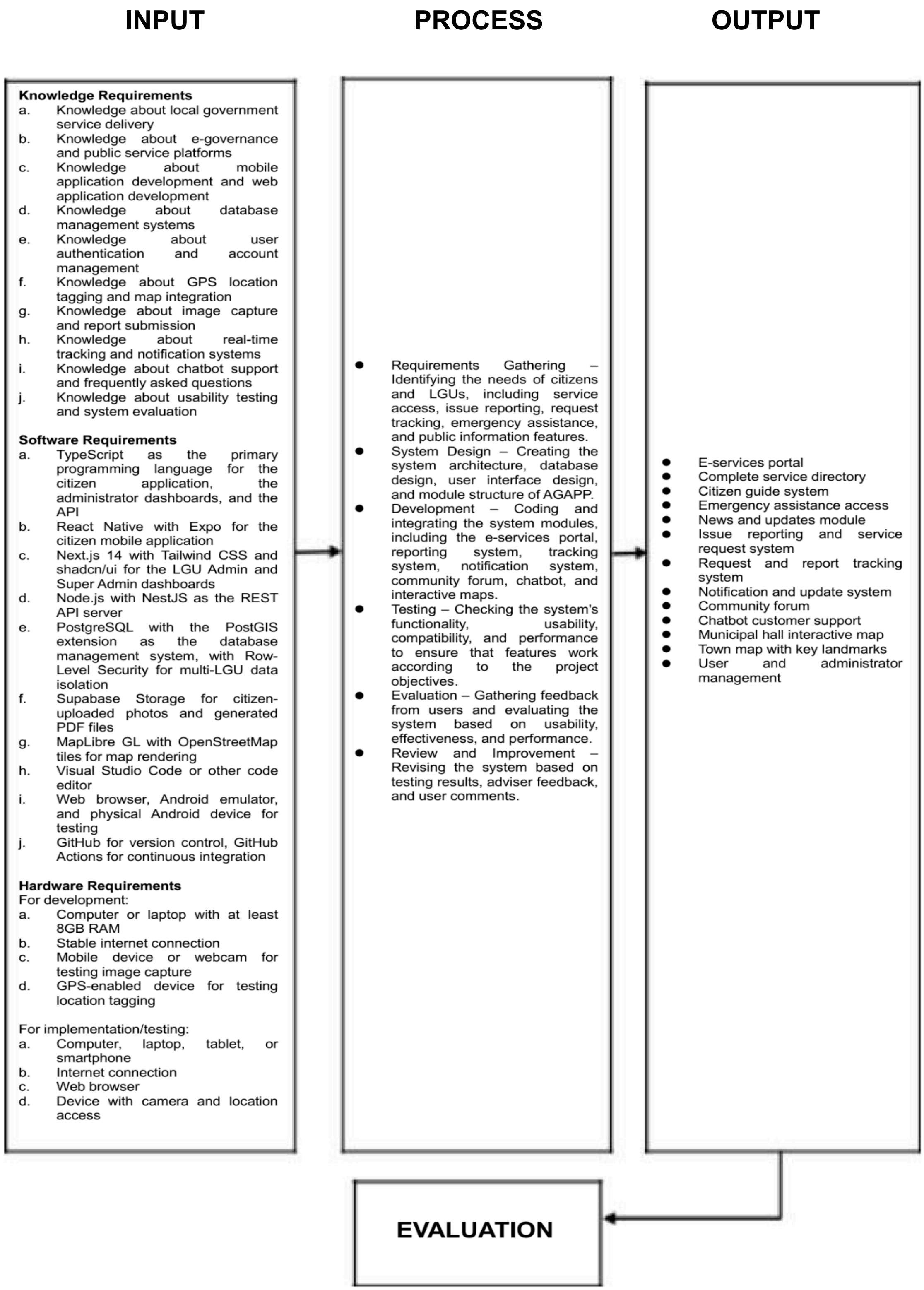

Figure 1. Conceptual Model of AGAPP: Automated Governance and Public Service Platform

The conceptual model for the project serves as a guide for the researchers to help identify the requirements and processes needed for the development of AGAPP. It consists of four stages: input, process, output, and evaluation. The input stage consists of knowledge requirements, software requirements, and hardware requirements, all of which help set the right development environment for the developers of the project. The knowledge requirements include understanding local government service delivery, e-governance platforms, mobile and web application development, database management systems, user authentication, GPS location tagging, image capture, real-time tracking, chatbot systems, and usability testing. The software requirements include TypeScript as the programming language, React Native with Expo for the mobile app, Next.js 14 for the admin dashboards, Node.js with NestJS for the API server, PostgreSQL with PostGIS for the database, Supabase Storage for file storage, MapLibre GL for maps, Visual Studio Code as the editor, and GitHub for version control. The hardware requirements include a computer with at least 8GB RAM, stable internet connection, Android device for testing, and GPS-enabled device for location testing.

The process stage shows the flow of the development of AGAPP from planning to completion. It includes requirements gathering to identify citizen and LGU needs, system design for architecture and database structure, development of all modules, testing for functionality and usability, evaluation through user feedback, and review and improvement based on results.

The output stage consists of the final AGAPP system with thirteen features: E-services portal, service directory, citizen guide system, emergency assistance access, news and updates module, issue reporting and service request system, request and report tracking system, notification and update system, community forum, chatbot customer support, municipal hall interactive map, town map with key landmarks, and user and administrator management.

The last stage consists of the overall evaluation, feedback, and results of the project. The evaluation assesses the system's functionality, usability, performance, and effectiveness to determine if AGAPP meets its objectives and improves accessibility, transparency, communication, and efficiency in local government service delivery.

The evaluation phase will assess the system's functionality, usability, performance, and effectiveness. This will help determine whether AGAPP meets its objectives and whether it improves accessibility, transparency, communication, and efficiency in local government service delivery.

Operational Definition of Terms

Administrator – Refers to the authorized user responsible for managing system records, reviewing reports, updating service information, and monitoring requests submitted through AGAPP.

AGAPP – Refers to the Automated Governance and Public Service Platform, the proposed centralized digital platform (composed of a citizen mobile application and web-based administrator dashboards) designed to improve local government service delivery.

Automated Governance – Refers to the use of digital tools and system features to organize, monitor, and improve public service processes.

Citizen Guide System – Refers to the feature that provides structured information about government services, procedures, requirements, schedules, and office locations.

Citizens/Residents – Refers to the users of the system who access services, submit requests, report issues, track concerns, and receive updates.

Community Forum – Refers to the feature where users can post concerns, share information, and participate in discussions related to their community.

E-Services Portal – Refers to the module that allows users to submit applications for permits and document requests online. It generates a pre-filled PDF and a reference QR code that the citizen presents at the Municipal Hall payment counter in person to complete payment and claim the released document.

Emergency Assistance Access – Refers to the feature that provides quick access to emergency hotlines such as police, fire protection, hospitals, and other emergency services.

GPS Location Tagging – Refers to the use of geographic location data to identify the exact location of a reported issue or submitted request.

Interactive Map – Refers to the digital map feature that displays municipal offices, service areas, key landmarks, and important town locations.

Issue Reporting System – Refers to the module that allows users to submit reports about community concerns such as potholes, clogged drainage, damaged utility poles, stray animals, missing pets, and lost or found items.

Local Government Unit (LGU) – Refers to the city or municipal government responsible for providing public services to citizens within a specific locality.

Notification System – Refers to the feature that sends alerts, announcements, advisories, and status updates to users.

Prototype – Refers to the preliminary version of AGAPP developed for demonstration, testing, and evaluation purposes.

Real-Time Tracking System – Refers to the feature that allows users to monitor the progress of submitted requests and reports using statuses such as Submitted, Under Review, In Progress, Resolved, and Rejected.

Service Directory – Refers to the list of available government services, including procedures, requirements, schedules, contact information, and office locations.

User Interface – Refers to the visual layout and design of the system that allows users to interact with AGAPP.

User Personalization – Refers to system settings that allow users to customize their experience, such as choosing between light mode and dark mode.